Homo Algebra by Rotman, Chapter 2

Contents

Simplex **1**

Correspondence Theorem **1**
Ring Version 1

Theorem 2.30 / 2.31 **1**

Extend by Linearity **1**

Exactness of Hom Functors **2**

$R/\langle r\rangle \otimes M \cong M/rM$ **for** $r \in Z(R)$ **2**

Adjoint Isomorphism **2**
Nautral Isos 2
Right Exactness of Tensor 3

Simplex

Let X be the punctured plane and Δ be the triangle with the deleted origin. Then there are lines which cannot form an area because they lie on both sides of the origin. Therefore Δ contains a 1-dimensional hole.

Correspondence Theorem

$$\varphi : \{\text{intermediate submodules } T \subseteq S \subseteq M\} \to \{\text{submodules of } M/T\}$$

and $T \subseteq S \subseteq S'$ in $M \iff S/T \subseteq S'/T$ in M/T.

Every module is an additive abelian group. Thus submodules are subgroups. So φ is an injection: $S \subseteq S'$ in $M \iff S/T \subseteq S'/T$ in M/T.

φ is surjective too: if $S^* \subseteq M/T$ then there's a submodule $S \supseteq T$ with $S^* = S/T$.

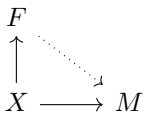
Ring Version

$$\varphi : \{\text{intermediate left ideals } I \subseteq J \subseteq R\} \to \{\text{left ideals of } R/I\}$$

Theorem 2.30 / 2.31

$$\varphi : \mathrm{Hom}(A, \prod B_i) \to \prod \mathrm{Hom}(A, B_i)$$
$$\psi : \mathrm{Hom}(\bigoplus A_i, B) \to \prod \mathrm{Hom}(A_i, B)$$

Extend by Linearity



Every $M \in {}_R\text{Mod}$ is a quotient of a free $F \in {}_R\text{Mod}$.

Exactness of Hom Functors

$$0 \to A \to B \to C \to 0$$

$\text{Hom}(X, -)$ is covariant right exact functor.

$$0 \to \text{Hom}(X, A) \to \text{Hom}(X, B) \to \text{Hom}(X, C)$$

$$0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$$

$\text{Hom}(\mathbb{Z}_2, \mathbb{Q}/\mathbb{Z}) \neq 0$ because it contains $f(1) = \frac{1}{2} + \mathbb{Z}$. But $\text{Hom}(\mathbb{Z}_2, \mathbb{Q}) = 0$ so that $p^* : \text{Hom}(\mathbb{Z}_2, \mathbb{Q}) \to \text{Hom}(\mathbb{Z}_2, \mathbb{Q}/\mathbb{Z})$ is not surjective.

$\text{Hom}(-, X)$ is contravariant left exact functor.

$$0 \to \text{Hom}(C, X) \to \text{Hom}(B, X) \to \text{Hom}(A, X)$$

Since $\text{Hom}(\mathbb{Q}, \mathbb{Z}) = 0$, the induced i^* cannot be surjective as $1 \in \text{Hom}(\mathbb{Z}, \mathbb{Z})$.

$R/\langle r \rangle \otimes M \cong M/rM$ for $r \in Z(R)$

For abelian groups this gives us $\mathbb{Z}_n \otimes G \cong G/nG$.

$$\begin{array}{ccccccc} R \otimes M & \longrightarrow & R \otimes M & \longrightarrow & R/\langle r \rangle \otimes M & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \vdots & & \\ M & \longrightarrow & M & \longrightarrow & M/rM & \longrightarrow & 0 \end{array}$$

Adjoint Isomorphism

$$\tau : \text{Hom}_S(A \otimes_R B, C) \to \text{Hom}_R(A, \text{Hom}_S(B, C))$$

let $f : A \otimes_R B \to C$

$$\tau : f \to \tau(f) \text{ where } \tau(f)_a(b) = f(a \otimes b)$$

τ is injective, since if $\tau(f)_a = 0$ then $0 = \tau(f)_a(b) = f(a \otimes b)$ for all a, b. Thus $f = 0$ since all generators of $f(A \otimes B)$ are 0.

τ is surjective. Defn $F : A \to \text{Hom}_S(B, C)$ and $\varphi : A \times B \to C$ by $\varphi(a, b) = F_a(b)$.

$$\begin{array}{ccc} A \times B & \longrightarrow & A \otimes_R B \\ & \searrow & \vdots \\ & & C \end{array}$$

$$\tau' : \text{Hom}_S(B \otimes_R A, C) \to \text{Hom}_R(A, \text{Hom}_S(B, C))$$

Nautral Isos

$$Hom_S(- \otimes_R B, C) \to Hom_R(-, Hom_S(B, C))$$
$$Hom_S(A \otimes_R -, C) \to Hom_R(A, Hom_S(-, C))$$
$$Hom_S(A \otimes_R B, -) \to Hom_R(A, Hom_S(B, -))$$
$$Hom_R(-, Hom_S(B, C)) \to Hom_S(- \otimes_S B, C)$$
$$Hom_R(-, Hom_S(B, C)) \to Hom_S(B \otimes_S -, C)$$

Right Exactness of Tensor

$$B' \to B \to B'' \to 0$$

Let $B \in {}_R\text{Mod}$ be a $(R, \mathbb{Z})$-bimodule. For any abelian group C, then $C \in \text{Mod}_{\mathbb{Z}}$ and so $\text{Hom}_{\mathbb{Z}}(B, C) \in \text{Mod}_R$.
Now given proposition 2.42 which states that if for every $M \in {}_R\text{Mod}$ then

$$0 \to \text{Hom}(B'', M) \to \text{Hom}(B, M) \to \text{Hom}(B', M)$$

then

$$B' \to B \to B'' \to 0$$

Therefore we can equivalently show that for every C, the top row is exact below

$$\begin{array}{ccccccc}
0 & \longrightarrow & \text{Hom}_{\mathbb{Z}}(A \otimes_R B'', C) & \longrightarrow & \text{Hom}_{\mathbb{Z}}(A \otimes_R B, C) & \longrightarrow & \text{Hom}_{\mathbb{Z}}(A \otimes_R B', C) \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \text{Hom}_R(A, \text{Hom}_{\mathbb{Z}}(B'', C)) & \longrightarrow & \text{Hom}_R(A, \text{Hom}_{\mathbb{Z}}(B, C)) & \longrightarrow & \text{Hom}_R(A, \text{Hom}(B', C))
\end{array}$$

The bottom row is exact by applying $Hom_R(A, Hom_{\mathbb{Z}}(-, C))$ to $B' \to B \to B'' \to 0$.
The vertical arrows are iso.
So the top row is exact.